

# The Behaviour of Gases

## *Chapter 11*



# Kinetic Molecular Theory of Gases

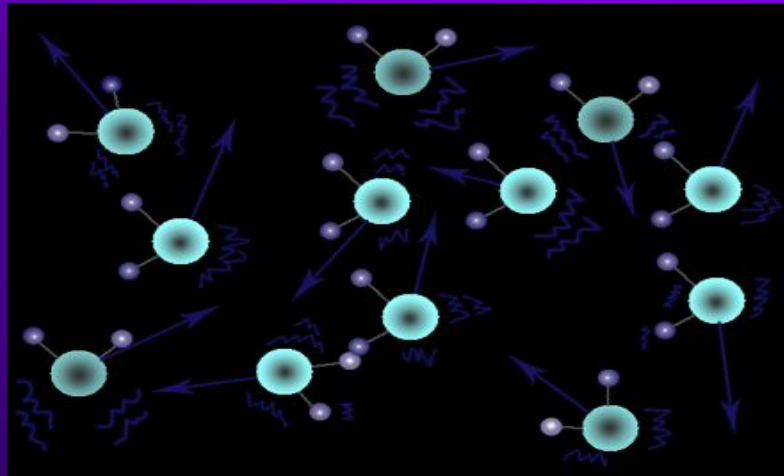
*This theory describes what we refer to as an “ideal gas”.*

- The volume of an individual gas molecule is negligible compared to the volume of the container holding the gas. Thus the molecules are very far apart and gases are considered compressible.
- There are neither attractive nor repulsive forces between gas molecules.
- Gas molecules have high translational energy.
- Collisions of gas molecules are completely elastic, thus when gas molecules collide there is no loss in kinetic energy.
- The average kinetic energy of gas molecules is directly proportional to temperature as well as mass and inversely proportional to pressure.

# States of Matter and the KMT

## Types of Motion:

1. Translational Motion – the independent motion of particles from one point in space to another
2. Rotational Motion – the motion of particles around other particles
3. Vibrational Motion – the back and forth motion of particles that are fixed in position



# States of Matter and the KMT

## Comparing Solids, Liquids & Gases:

| <b>SOLIDS</b>                                            | <b>LIQUIDS</b>                                        | <b>GASES</b>                                                             |
|----------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Incompressible<br/>(definite volume)</b>              | <b>Incompressible<br/>(definite volume)</b>           | <b>Compressible<br/>(indefinite volume)</b>                              |
| <b>Crystal lattice<br/>(fixed structure)</b>             | <b>Indefinite shape</b>                               | <b>Indefinite shape</b>                                                  |
| <b>Cannot flow</b>                                       | <b>Flows<br/>(According to gravity)</b>               | <b>Flows<br/>(In all directions)</b>                                     |
| <b>Vibrational Motion</b>                                | <b>Vibrational Motion<br/>Rotational Motion</b>       | <b>Vibrational Motion<br/>Rotational Motion<br/>Translational Motion</b> |
| <b>Very low<br/>Degree of Disorder</b>                   | <b>Medium<br/>Degree of Disorder</b>                  | <b>Very High<br/>Degree of Disorder</b>                                  |
| <b>Strongest attractive<br/>forces between particles</b> | <b>Strong attractive forces<br/>between particles</b> | <b>Weakest attractive forces<br/>between particles</b>                   |

# States of Matter and the KMT

## Intermolecular Forces:

Forces between neutral molecules or between molecules & ions

- **Dipole-dipole Forces** – the positive pole of one polar molecule is attracted to the negative pole of another polar molecule.
- **Weak Dispersion Forces** – a temporary force of attraction between non-polar molecules

**Temperature:** the average kinetic energy of the particles in a substance

**Kinetic energy:** the energy of motion

# HOMEWORK

- Read text p.416-423
- -make notes by following questions on gas handout package
- (Kinetic Molecular Theory = KMT)
- -Text Questions p.423 #1-8

# GAS LAWS

# Introduction to Gas Laws

*The gas laws look at four factors:*

1. **Volume (L)**
  - Volume of a container
2. **Temperature (K)**
  - Measure of kinetic energy
3. **Pressure (kPa)**
  - $P = F/A$  (N/m<sup>2</sup>)
4. **Mass (g)**

## The Law of Proportionality

If  $a \propto b$  we will introduce a constant “k”

Then  $a = kb$



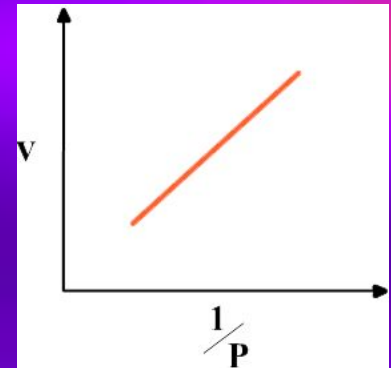
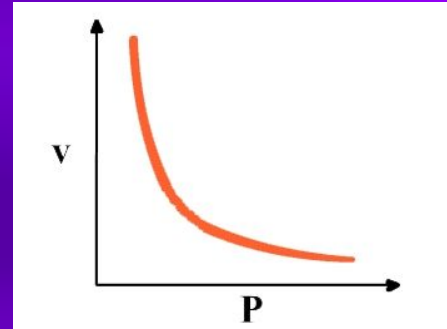
# Boyle's Law

The volume of a fixed mass of gas varies *inversely* with the applied pressure when the temperature is kept constant.

$$V \propto \frac{1}{P} \text{ for } T = \text{constant}$$

Thus

$$V = k \left( \frac{1}{P} \right)$$



We can rearrange this formula to yield

$$PV = k$$

Since  $k$  is a constant, it is true for a gas at any pressure!

Thus for case 1 (initial conditions) and case 2 (final conditions)

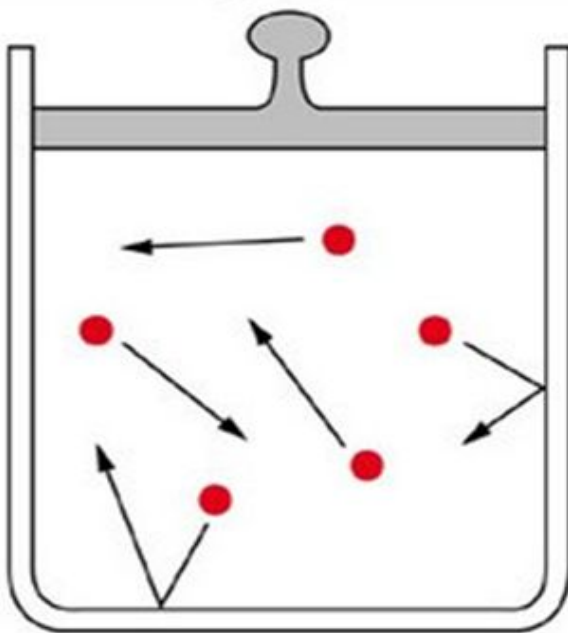
$$P_1 V_1 = P_2 V_2$$

# Boyle's Law

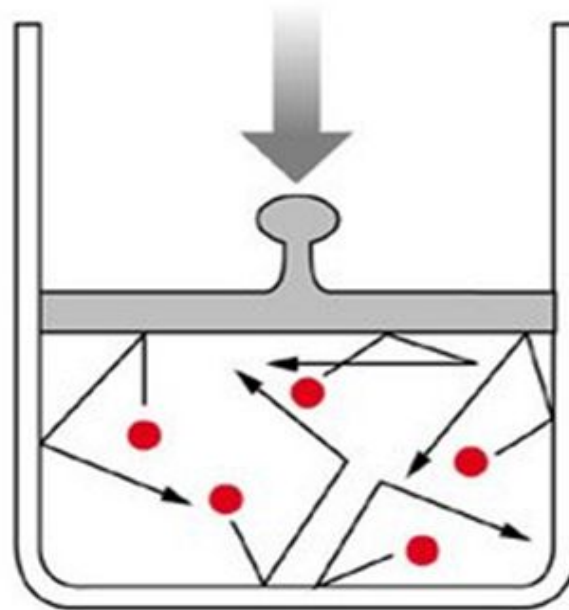
$$P_1 V_1 = P_2 V_2$$

**Boyle's Law:  $P_1 V_1 = P_2 V_2$**

**Decreasing volume increases collisions and increases pressure.**



$V_1 = 1.0 \text{ L}$   
 $P_1 = 100 \text{ mm Hg}$



$V_2 = 0.5 \text{ L}$   
 $P_2 = 200 \text{ mm Hg}$

# Boyle's Law

- *The Units!*

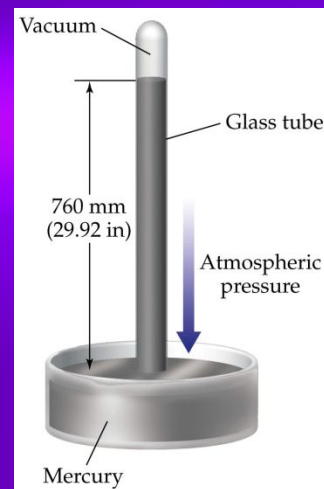
Volume must be measured in litres (L) or cubic centimetres ( $\text{cm}^3$ ) and pressure usually in kilopascals (kPa), but other units are accepted.

Standard atmospheric pressure at  $0^\circ\text{C}$   
is 101.3 kPa

Numerous units may be used for pressure

|       |                        |
|-------|------------------------|
| atm   | atmospheres            |
| mm Hg | millimeters of mercury |
| torr  | torr                   |
| psi   | pounds per square inch |

$$101.3 \text{ kPa} = 1 \text{ atm} = 760 \text{ mm Hg} \\ = 760 \text{ torr} = 14.7 \text{ psi}$$



# Boyle's Law

## ***Sample Problem:***

A 4.50 L balloon filled with helium is collected at room temperature and standard atmospheric pressure. The balloon is then submerged in a tub of water, also at room temperature such that the external pressure is increased to 110.2 kPa.

What is the final volume of the balloon?

## ***Practice Problems:***

p.434 #1-4

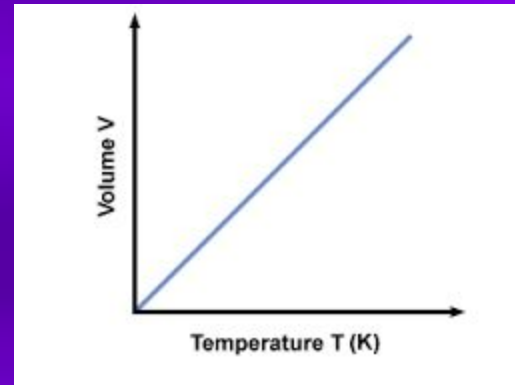
p.435 #1-6

# Charles' Law

The volume of a fixed mass of gas is *proportional* to its temperature when the pressure is kept constant.

$$V \propto T$$

$$\text{Thus } V = kT$$

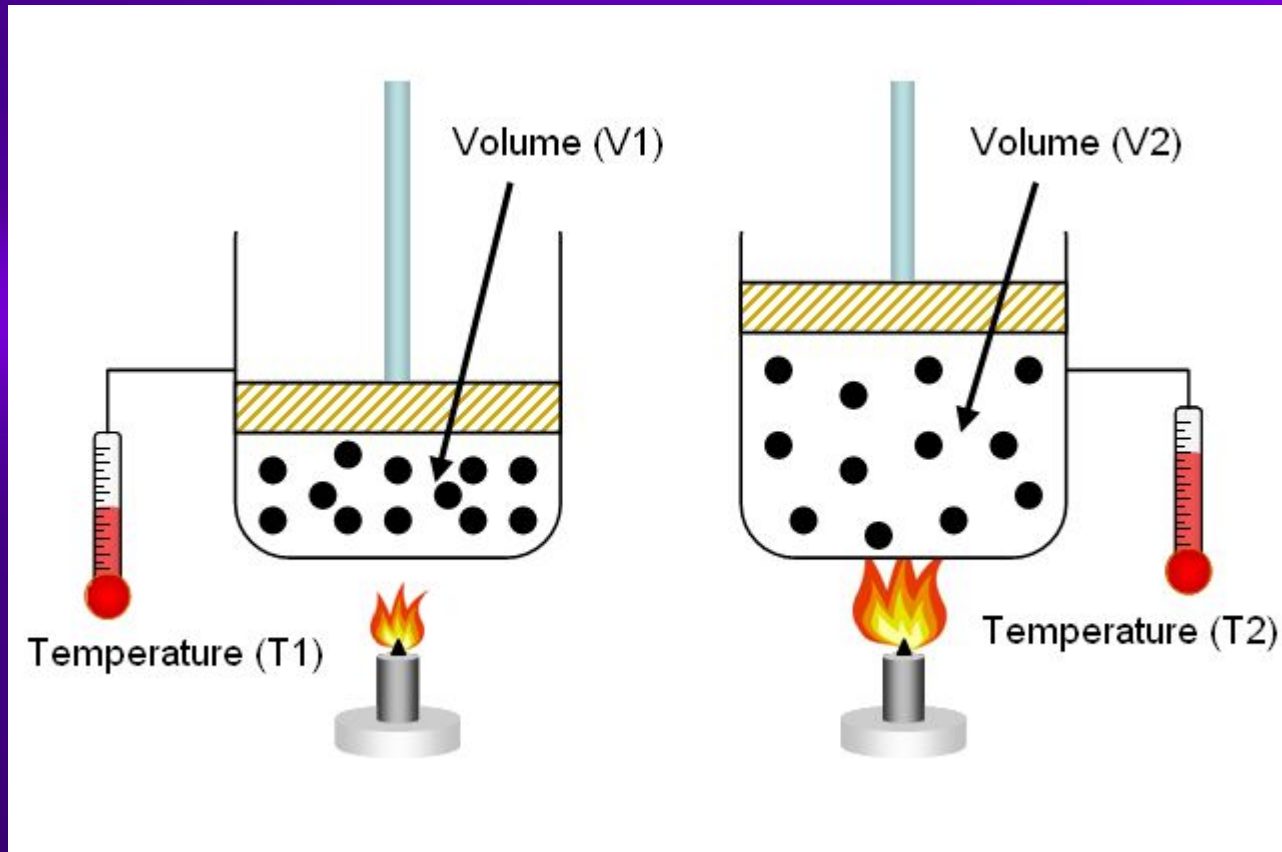


- We can rearrange this formula to yield  $k = V/T$
- Since  $k$  is a constant it is true for a gas at any temperature!
- Thus for case 1 (initial conditions) and case 2 (final conditions)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

# Charles' Law

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



# Charles' Law

## *There is a catch!*

- Temperature must be measured in Kelvin and volume in litres
- The Kelvin scale is similar to the Celsius scale (in graduations)
- To determine the Kelvin temperature just add 273.15 to the Celsius scale (we will use 273)

$$T_K = T_{^{\circ}\text{C}} + 273$$

- Unlike other temperature scales K has no  $^{\circ}$  symbol.

|               |       |       |
|---------------|-------|-------|
| B.P. of water | 100°C | 373 K |
|---------------|-------|-------|

|            |      |       |
|------------|------|-------|
| Body Temp. | 37°C | 310 K |
|------------|------|-------|

|               |     |       |
|---------------|-----|-------|
| F.P. of water | 0°C | 273 K |
|---------------|-----|-------|

|               |        |     |
|---------------|--------|-----|
| Absolute Zero | -273°C | 0 K |
|---------------|--------|-----|

- Present and Previous Names for these scales:

Celsius    ☐    Centigrade

Kelvin    ☐    Absolute



# Gay-Lussac's Law

The pressure of a fixed mass of gas varies *directly* with the temperature when the volume is kept constant.

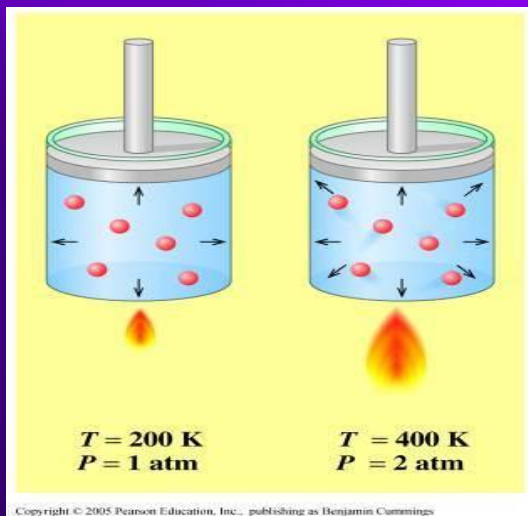
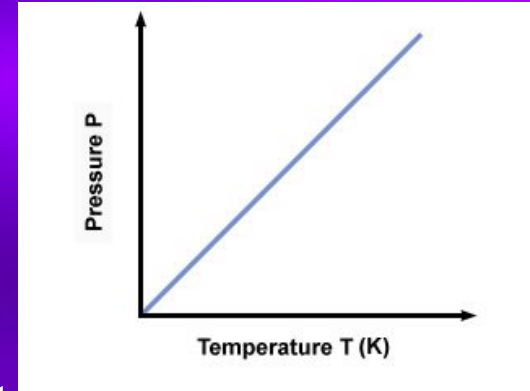
$$P \propto T$$

Thus  $P = kT$

We can rearrange this formula to yield  $k = P/T$

Since  $k$  is a constant it is true for a gas at any temperature!

Thus for case 1 (initial conditions) and case 2 (final conditions):



$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$



# Gay-Lussac's Law

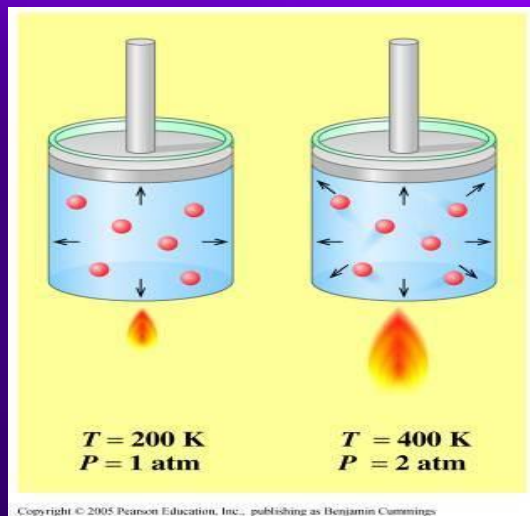
•*There is a catch!*

Temperature must be measured in Kelvin (K) and pressure usually in kilopascals (kPa), but other units are accepted.

Recall:

101.3 kPa = 1 atm = 760 mmHg = 760 torr = 14.7 psi

**STP 101.3 kPa & 0° C & 273 K**



$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

## ***Sample Problems:***

1. A balloon is filled with helium at standard atmospheric pressure and  $22.5^{\circ}\text{C}$ . When I took the balloon outside, the pressure was 90.1 kPa. What was the outside temperature in  $^{\circ}\text{C}$ ?
2. The pressure gauge reading on a cylinder of oxygen gas is  $8.5 \times 10^3$  kPa, at  $22^{\circ}\text{C}$ . What will be the pressure if the cylinder is placed in boiling water?

## ***Practice problems:***

p.446 #5,6,8-12

p.449 #13 – 16

p.451 #1-5

# Combining The Gas Laws...

- Look at all the gas laws we have considered...
- Can you create a relationship that accounts for all of them in one formula?
- Work with a partner and try to figure it out...

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- 2 new “**standard conditions**”
- STP – 0°C and 101.3 kPa (at sea level)
- SATP – standard ambient temperature & pressure 25°C and 100 kPa (room conditions)

## *Sample problems...*

1. For my son's birthday, a party store filled a 745 mL balloon with helium at SATP. When I took the balloon outside, the temperature was  $-9.4\text{ }^{\circ}\text{C}$  and the atmospheric pressure was 99.7 kPa. What was the volume of the helium outside?
2. To what temperature (in  $^{\circ}\text{C}$ ) must 25.0 mL of argon gas at STP be heated such that the new volume is 30.0 mL and the pressure is 110 kPa?

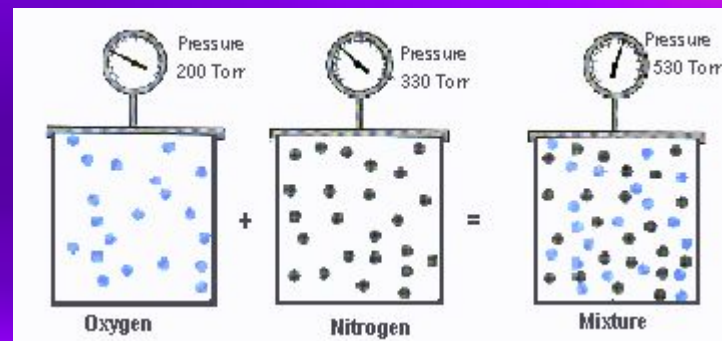
**HW:**

p.457 #17-21

# Dalton's Law of Partial Pressures

- The total pressure of a mixture of gases is the sum of the pressures of each of the individual gases

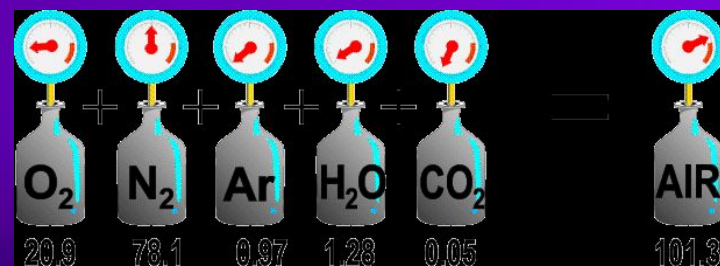
$$P_T = P_1 + P_2 + P_3 + \dots$$



## Sample Problem:

What is the pressure contribution of  $\text{CO}_2$  to the atmospheric pressure on a very dry day when the barometer reads 0.98 atm?

□ use table 11.2 p.459

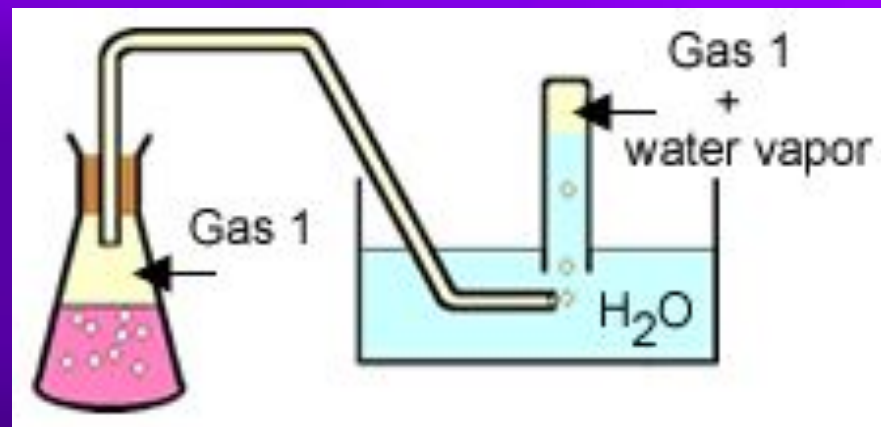


# Dalton's Law of Partial Pressures

*DEMO...*

$$P_T = P_1 + P_2 + P_3 + \dots$$

- When gas is collected over water, , molecules of water vapour mix with the gas sample.
- We must subtract the pressure that the water vapour contributes from the total pressure in order to determine the pressure of the gas collected.



# Dalton's Law of Partial Pressures

## *Sample problem:*

$$P_T = P_1 + P_2 + P_3 + \dots$$

A volume of 110 mL of  $H_2(g)$  is collected over water at a temperature of  $17^\circ\text{C}$  and an atmospheric pressure of 95.0 kPa. What is the volume of dry hydrogen at STP?

– See table 12.3 p.507 pressure of water vapour

$$P_T = P_{H_2} + P_{H_2O}$$

$$\begin{aligned} P_{H_2} &= P_T - P_{H_2O} \\ &= 95.0 \text{ kPa} - 1.94 \text{ kPa} \\ &= 93.1 \text{ kPa} \end{aligned}$$

$$V_1 = 110 \text{ mL}$$

$$T_1 = 17^\circ\text{C} = 290 \text{ K}$$

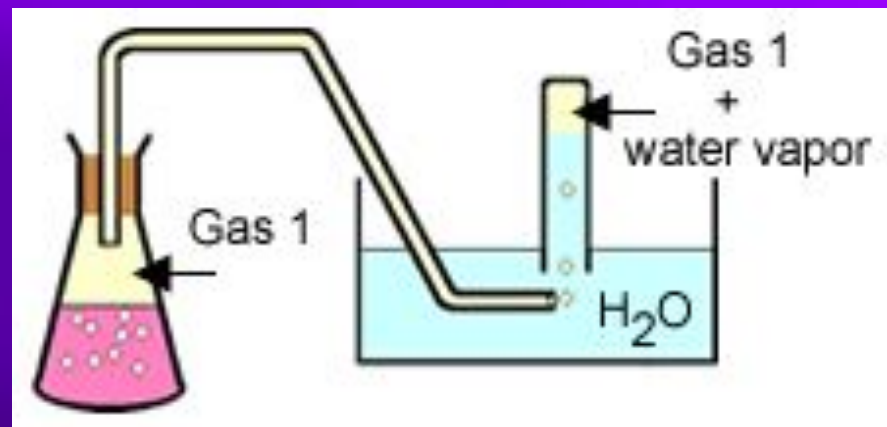
$$P_2 = 101.3 \text{ kPa}$$

$$T_2 = 273 \text{ K}$$

$$V_2 = ?$$

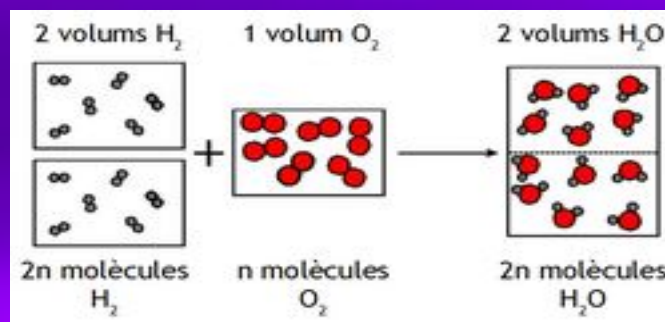
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

**HW: complete  
Q#1-4 on  
handout**





- Our friend Gay-Lussac was confused!
- When he combined 2 volumes of hydrogen and 1 volume of oxygen, he got only two volumes of water vapour. What was going on? (see p.472 – 474)



- Gay-Lussac did not know that gas molecules could contain more than one atom!
- He consulted with Dalton (law of multiple proportions) and they published their results.
- **Avogadro figured it out!**

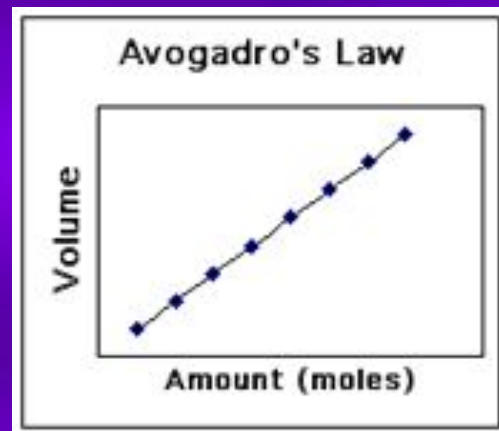


# Avogadro's Hypothesis

Equal volumes of gases at the same temperature and pressure contain the same number of molecules.

$$V \propto n$$

Thus  $V = kn$



We can rearrange this formula to yield  $k = V/n$

Since  $k$  is constant it is true for a gas at any pressure!

Thus for case 1 (initial conditions) and case 2 (final conditions)

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

## •The Units!

Volume must be measured in litres (L) or cubic centimetres ( $\text{cm}^3$ )

# Avogadro's Hypothesis

## ***Sample Problem:***

1. For my son's birthday, a party store filled a 745 mL birthday balloon with 3.2 g helium @ SATP. When I took the balloon home it had a slow leak and lost 0.5 g helium, the temperature was 25 °C and the atmospheric pressure was 100 kPa. What was the volume of the balloon at home?

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

# Avogadro's Hypothesis

## ***Sample Problem:***

2. 1.30 L of  $\text{N}_{2(g)}$  at  $22.0\text{ }^{\circ}\text{C}$  and  $98.0\text{ kPa}$  have a mass of  $1.45\text{ g}$ . Use this information to calculate the molar volume of  $\text{N}_{2(g)}$  at STP.

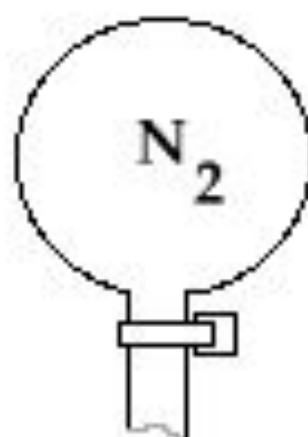
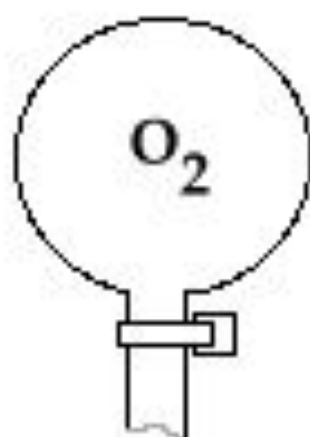
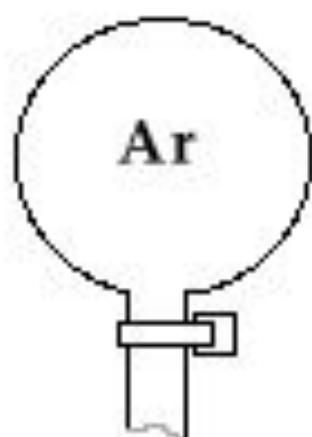
**Note:** Molar volume is the volume occupied by one mole of a gas.

This is the same for ANY gas at STP **22.4 L**

At SATP, the molar volume of any gas is **24.8 L**

***HW: p.477 #1,3 & 4***  
***p.482 #5,6,6,10***

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$



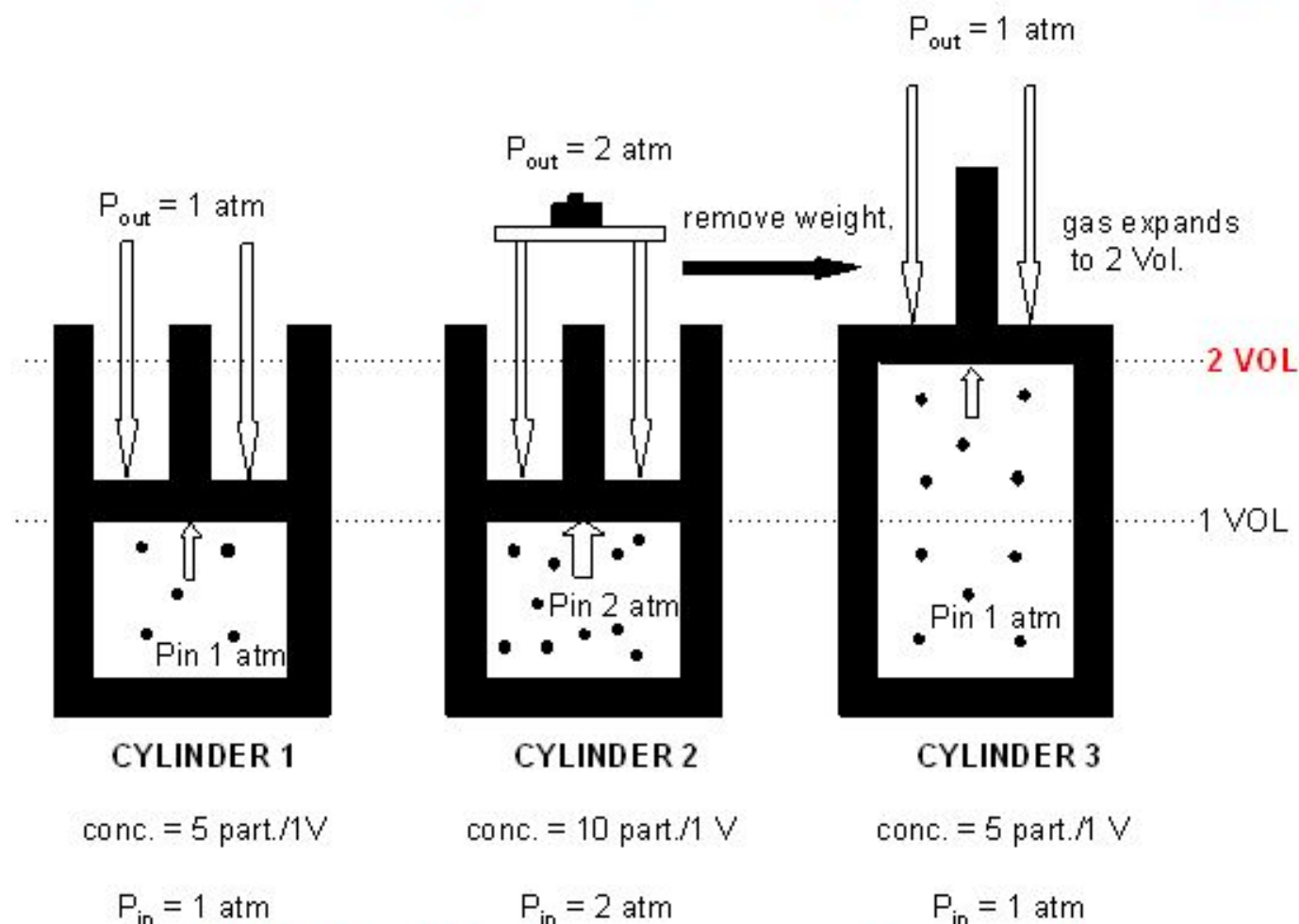
Volume: 22.4 L  
Mass: 40 g  
Quantity: 1 mol  
Pressure: 1 atm  
Temperature: 273 K

22.4 L  
32 g  
1 mol  
1 atm  
273 K

22.4 L  
28 g  
1 mol  
1 atm  
273 K

# AVOGADRO'S HYPOTHESIS

Equal Volumes of gases at equal temp. and press. have the same number of particles.  
Particles are molecules for most gases, excepts for those gas that exist as atoms (e.g. He).



**Cylinder 1 and 3 have the same pressure and temperature!**

Gas particles inside cylinder striking the interior walls  
and bottom surface of piston cause internal pressure

What if we consider Avagadro's law  
too...

the IDEAL Gas Law

$$PV=nRT$$

# Ideal Gas Law

How did we get there??

- Recall: (for constant T & P)

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- And (for a fixed amount of gas)

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- Therefore

$$\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$$

- This means

$$\frac{PV}{nT} = R$$

Where: **R = Universal Gas Law Constant**



# Ideal Gas Law

The value of R can be calculated using one mole of a gas at STP:

$$P = 101.3 \text{ kPa}$$

$$V = 22.4 \text{ L}$$

$$T = 273 \text{ K}$$

$$n = 1.00 \text{ mol}$$

$$\frac{PV}{nT} = R$$

$$R = 8.314 \frac{\text{kPa L}}{\text{mol K}}$$

R can also be expressed in other units:

$$R = 0.08206 \frac{\text{atm L}}{\text{mol K}}$$

$$R = 62.37 \frac{\text{mmHg L}}{\text{mol K}}$$

The Ideal Gas Law Equation can be rearranged as:

$$PV = nRT$$

# Ideal Gas Law

## *Sample Problems:*

1. The pressure gauge on a 15.0 L cylinder of oxygen gas has a reading of  $2.70 \times 10^3$  kPa and a temperature of  $22.5^\circ\text{C}$ . What is the mass of oxygen in the cylinder?
2. What volume of carbon dioxide did a swimmer breathe out when he exhaled 10.5 g of  $\text{CO}_2$  @ SATP into a flask to measure his lung capacity?

## HW:

-p.487 # 12-14

-p.488 #1-5

$$PV=nRT$$

Pressure



Temperature



Number of moles



$$PV=nRT$$

Volume



Gas constant

